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## SQL Window Functions

### 1. Employee sequence by salary — ROW\_NUMBER()

```
SELECT emp_name, department, salary,  
       ROW_NUMBER() OVER (ORDER BY salary DESC) AS row_num  
FROM employees;
```

### 2. Salary rank across company — RANK()

```
SELECT emp_name, department, salary,  
       RANK() OVER (ORDER BY salary DESC) AS salary_rank  
FROM employees;
```

### 3. Salary rank without gaps — DENSE\_RANK()

```
SELECT emp_name, department, salary,  
       DENSE_RANK() OVER (ORDER BY salary DESC) AS salary_rank  
FROM employees;
```

### 4. Row number inside each department — ROW\_NUMBER()

```
SELECT emp_name, department, salary,  
       ROW_NUMBER() OVER (  
         PARTITION BY department  
         ORDER BY salary DESC  
       ) AS row_num  
FROM employees;
```

### 5. Department-wise salary rank — RANK()

```
SELECT emp_name, department, salary,  
       RANK() OVER (  
         PARTITION BY department  
         ORDER BY salary DESC  
       ) AS salary_rank  
FROM employees;
```

### 6. Top 2 distinct salaries in each department — DENSE\_RANK()

Use a subquery:

```
SELECT emp_name, department, salary
```

```
FROM (
```

```
    SELECT emp_name,
```

```
        department,
```

```
        salary,
```

```
        DENSE_RANK() OVER (
```

```
            PARTITION BY department
```

```
            ORDER BY salary DESC
```

```
        ) AS rnk
```

```
    FROM employees
```

```
) AS temp
```

```
WHERE rnk <= 2;
```

### **7. Highest-paid employee name per department — FIRST\_VALUE()**

```
SELECT emp_name,
```

```
    department,
```

```
    salary,
```

```
    FIRST_VALUE(emp_name) OVER (
```

```
        PARTITION BY department
```

```
        ORDER BY salary DESC
```

```
    ) AS highest_paid_employee
```

```
FROM employees;
```

### **8. Lowest-paid employee name per department — LAST\_VALUE()**

This is where the **window frame** is important.

```
SELECT emp_name,
```

```
    department,
```

```
    salary,
```

```
    LAST_VALUE(emp_name) OVER (
```

```
        PARTITION BY department
```

```
        ORDER BY salary DESC
```

```
        ROWS BETWEEN UNBOUNDED PRECEDING AND UNBOUNDED FOLLOWING
```

```
    ) AS lowest_paid_employee
```

FROM employees;

### 9. Highest and lowest salary in each department

SELECT emp\_name,

department,

salary,

FIRST\_VALUE(salary) OVER (

PARTITION BY department

ORDER BY salary DESC

) AS highest\_salary,

LAST\_VALUE(salary) OVER (

PARTITION BY department

ORDER BY salary DESC

ROWS BETWEEN UNBOUNDED PRECEDING AND UNBOUNDED FOLLOWING

) AS lowest\_salary

FROM employees;

### 10. 2nd highest distinct salary — DENSE\_RANK()

Use a subquery:

SELECT emp\_name, department, salary

FROM (

SELECT emp\_name,

department,

salary,

DENSE\_RANK() OVER (

ORDER BY salary DESC

) AS rnk

FROM employees

) AS temp

WHERE rnk = 2;